Survival Wiki

Field Guide

A printed copy of every article on the device. Keep it with the device as a backup for when batteries run out.

This is a field reference, not a substitute for training or professional medical care.

First Steps

Priorities, mindset and what to do first

The Rule of 3s

In an emergency, deal with the biggest threat first. The Rule of 3s is a rough guide to how long a person can last without each basic need.



The Rule of 3s

The order

1. **Air** - about 3 minutes. Breathing problems, choking, drowning, smoke and severe bleeding come first.
2. **Shelter** - about 3 hours in harsh weather. Cold, wet and wind kill faster than thirst. Stay dry and insulated.
3. **Water** - about 3 days. Less in heat or with hard work.
4. **Food** - about 3 weeks. Food matters for morale and energy, but it is the last priority.

> These are averages, not promises. Cold water, heat, injury, age and illness shorten every limit.

Using it

- Before you build anything, check for immediate danger: fire, rising water, traffic, falling rocks.

- Treat life-threatening injuries next.
- Then protect yourself from the weather, then find water.
- Spend energy on food only once the first three are handled.

See also: STOP: Stop, Think, Observe, Plan

STOP - Stop, Think, Observe, Plan

Most survival deaths come from bad decisions made in the first hour. When things go wrong, use **STOP**.

S - Stop

Sit down. Drink some water, eat something small. Do not keep walking. Panic burns energy and makes you more lost.

T - Think

- What happened, and what do you actually have?
- Does anyone know where you are and when you are due back?
- What is the biggest danger right now?

O - Observe

- Weather: is it getting colder, darker, wetter?
- Terrain: water sources, shelter material, open areas for signaling.
- Your body: injuries, warmth, thirst, how much daylight is left.
- Your gear: empty your pockets and pack. Everything counts.

P - Plan

Decide on one clear plan and follow it. Re-check it every few hours.

> If someone knows your route and expected return, it is usually best to **stay put**. Searchers find people who stay in one place far more often.

Stay or go?

Stay if: people will look for you, you are injured, the weather is bad, or you do not know which way to go.

Move only if: no one will search for you, your location is unsafe, or you know a clear route to help. If you move, leave a note or arrow showing your direction.

Basic Survival Kit

A small kit you actually carry beats a big kit left at home. Build it around the Rule of 3s.

Core items

- **Fire:** lighter, ferro rod, waterproof matches, tinder (cotton balls with petroleum jelly)
- **Shelter:** emergency blanket or bivy bag, small tarp, 15 m of cord
- **Water:** metal bottle you can boil in, purification tablets, filter
- **Light:** headlamp with spare batteries
- **Signal:** whistle, signal mirror, bright cloth
- **Tools:** fixed-blade or folding knife, duct tape
- **First aid:** bandages, gauze, tape, tourniquet, pain relievers, personal medicines
- **Navigation:** map of the area, compass
- **Food:** high-calorie bars

Clothing

- Warm layer (wool or synthetic, not cotton)
- Rain jacket
- Hat and gloves
- Spare socks

> Cotton loses its warmth when wet. In cold or wet weather, wool and synthetics can be the difference between cold and hypothermic.

Tell someone

The most important item is not in the kit: leave your route and expected return time with someone you trust.

Lightning and Weather Dangers

Lightning

- If you can hear thunder, you are close enough to be struck. Get to a building or hard-topped vehicle if you can.
- Avoid summits, ridges, open fields, lone trees, water and metal fences.
- In a forest, go to a low area among smaller trees of similar height.
- Groups should spread out at least 15 m (50 ft) apart so one strike does not hit everyone.
- Wait 30 minutes after the last thunder before going back out.

> There is no truly safe place outdoors in a thunderstorm. If caught in the open with no option, crouch low on the balls of your feet, feet together, away from tall objects.

Flash floods

- Never camp in a dry stream bed or narrow canyon.
- Rain far upstream can cause a flood where the sky is clear.
- Do not walk or drive through moving water. 15 cm (6 in) of fast water can knock you down; 60 cm (2 ft) can float a car.
- Climb to higher ground if you hear a roar or see water rising and muddying.

Wind and cold fronts

- A sudden temperature drop, shifting wind and dark clouds mean bad weather is near. Get shelter set up early.
- Wind steals body heat fast. Block the wind before anything else.

Essential Knots

Learn a few knots well rather than many badly. Practice them until you can tie them with cold hands or in the dark.

Bowline - a loop that will not slip

Use it for a fixed loop: tying a line around a tree, a rescue loop, hanging a food bag.

1. Make a small loop in the standing part of the rope, with the working end lying on top.
2. Pass the working end **up through** the small loop from underneath.
3. Take it **around behind** the standing part.
4. Bring it back **down through** the small loop.
5. Hold the working end against the big loop and pull the standing part to tighten.

Leave a tail at least 10 cm (4 in) long. A bowline can loosen when it is not under load, so check it.

Clove hitch - quick tie to a pole

Use it to start lashings or tie a line to a post.

1. Wrap the rope once around the pole.
2. Cross over the first wrap and go around again.
3. Tuck the end under the second wrap, so both ends come out between the wraps in opposite directions.
4. Pull both ends tight.

Taut-line hitch - an adjustable tent line

Lets you tighten or loosen a guy line without retying.

1. Pass the rope around a stake or anchor.
2. Wrap the working end **twice** around the standing part, working back toward the anchor, inside the loop.
3. Make **one more wrap** around the standing part on the outside of those two wraps.
4. Tighten. Slide the knot to adjust the tension. It holds under load.

Square (reef) knot - tie two ends together

For bandages, bundles and packages.

1. Right end over left, and tuck under.

2. Left end over right, and tuck under.
3. Both ends should lie flat, alongside their own parts.

> Do not use a square knot to join two ropes that will carry weight. It can slip or spill. Use a sheet bend.

Sheet bend - join two ropes

Works even with ropes of different thickness.

1. Make a bend (U shape) in the end of the thicker rope.
2. Pass the thinner rope **up through** the bend from behind.
3. Wrap it **around behind** both sides of the bend.
4. Tuck it **under itself**, not back through the bend. Pull tight.

Trucker's hitch - pull a line very tight

Great for ridgelines and tying down loads.

1. Tie one end to an anchor.
2. Partway along the line, twist a small loop into the rope (a slip loop works).
3. Run the free end around the second anchor and back up through that loop.
4. Pull down on the free end. The loop acts like a pulley and gives you about twice the pull.
5. Hold the tension and lock it off with two half hitches below the loop.

Rope care

- Melt or tape the ends of synthetic cord so they do not fray.
- Keep rope out of fire and away from sharp edges.
- Paracord can be pulled apart: the inner strands make fishing line, sewing thread or snare line.

Dangerous Animal Encounters

Most wild animals avoid people. Encounters usually happen when an animal is surprised, defending young or food, or used to human food.

Avoid them

- Make noise in thick brush and near streams so animals hear you coming.
- Never leave food, trash or toiletries in your shelter. See Keeping Food and Camp Safe.
- Travel in groups during the day. Keep children close.
- Give all animals space, especially mothers with young.
- Carry **bear spray** in bear country, where you can reach it in a second.

Bears - general rules

- **Do not run.** Bears are much faster than people, and running can trigger a chase.
- Stay calm and talk in a low, firm voice so it knows you are human.
- Pick up small children. Stand together as a group.
- Back away slowly, sideways, while watching the bear.
- If it approaches, use bear spray when it is about 10 m (30 ft) away: aim slightly down so the cloud is between you and the bear.

If a bear attacks

Defensive attack (surprised bear, often a grizzly or brown bear, or a mother with cubs):

- Play dead. Lie flat on your stomach, hands clasped behind your neck, legs apart so it is hard to flip you. Keep your pack on.
- Stay still until the bear has left the area.
- If it keeps attacking for a long time, fight back hard.

Predatory attack (a bear stalks you, follows you, or attacks at night in your tent - often a black bear):

- **Fight back** with everything you have. Hit the face and nose with rocks, sticks, anything.
- Do not play dead.

Cougars (mountain lions)

- Do not run, and do not turn your back.

- Make yourself look big: raise your arms, open your jacket, stand on a rock.
- Shout, throw things, and back away slowly.
- If attacked, **fight back** and protect your neck. Stay on your feet if you can.

Moose and other large plant-eaters

Moose injure more people than bears in some regions. If ears go back and hair on its hump stands up, it is angry. Run and get behind a tree or vehicle: moose usually stop chasing quickly.

Dogs

- Do not run or stare. Stand still, side-on, hands close to your body.
- Say "No" or "Go home" firmly.
- If attacked, put something between you and the dog (a pack, jacket or stick). If knocked down, curl into a ball and protect your face and neck.

> Any animal bite or scratch that breaks the skin needs thorough washing for 15 minutes with soap and water, and medical care as soon as possible because of rabies and infection.

Ice Safety and Falling Through Ice

Is the ice safe?

Clear, hard, new ice is strongest. White or cloudy ice is weaker, often about half as strong.

- Under 5 cm (2 in): **stay off**.
- 10 cm (4 in) of clear ice: walking for one person.
- 13-18 cm (5-7 in): a snowmobile or small all-terrain vehicle.
- 20-30 cm (8-12 in): a small car.

Ice is never the same thickness everywhere. It is thinner near inlets, outlets, moving water, docks, rocks, logs and dark spots, and under snow. Check thickness every few steps with an ice chisel, axe or auger.

Never go onto ice alone. Carry ice picks (two spikes on a cord around your neck) and a throw rope.

If you fall in: the 1-10-1 rule



The 1-10-1 rule for falling through ice

- **1 minute:** cold shock makes you gasp and breathe fast. Keep your head above water, do not panic, and slow your breathing. It passes.
- **10 minutes:** of useful movement before your arms and legs go numb. Use them to get out now.
- **1 hour:** before you lose consciousness from hypothermia, even if you cannot get out. Keep your arms on the ice; they may freeze to it and hold you up.

Getting out

1. Turn toward the direction you came from. That ice held you before.
2. Put your arms flat on the ice. Use ice picks, keys or a knife to grip.
3. **Kick your legs hard** to bring your body up level with the surface, like a swimmer.
4. Pull and kick until your chest and hips are on the ice.
5. **Roll away** from the hole. Do not stand up until you are on thick ice.

6 Get to shelter and warmth immediately. See Hypothermia and Frostbite.

Helping someone else

Reach, throw, don't go. Do not run out to them; you will likely fall in too.

- Call for help.
- Reach with a long pole, branch, rope or ladder while lying flat or standing on solid ice or shore.
- Throw a rope or anything that floats.
- If you must go out, lie flat to spread your weight and push something ahead of you for them to grab.

Staying Calm and Keeping Morale

Survival is mostly mental. People with very little gear have survived because they kept their heads. People with plenty have died because they panicked.

Calm yourself in a minute

Box breathing: breathe in for 4 seconds, hold for 4, breathe out for 4, hold for 4. Repeat four or five times. It slows your heart and helps you think.

Then use STOP: Stop, Think, Observe, Plan.

Keep going

- **Give yourself small goals.** "Get the shelter roof done before dark" is easier than "survive."
- **Keep a routine:** set times to check signals, collect water and firewood, eat and sleep.
- **Stay busy.** Useful work keeps fear down.
- **Look after your body.** Cold, thirst, hunger and lack of sleep all make your thinking worse. Fix them early.
- **Keep a log.** Write the date, what you did, and what you plan. It helps you think, and helps rescuers later.

In a group

- Choose one leader to make final decisions, and listen to everyone first.

- Give every person a job, even children and injured people. Feeling useful matters.
- Check on each other. Watch for anyone who goes quiet, stops eating or wanders off.
- Keep the group together. Never let anyone leave camp alone without saying where and when they will be back.

With children

- Stay calm in front of them. They take their cues from you.
- Explain simply and honestly what is happening and what the plan is.
- Give them simple tasks and praise.
- Teach them: **if you are lost, stay put, and hug a tree.** Make noise, answer anyone calling your name, and do not hide from searchers.

> Rescue usually comes. Most people who are reported missing are found within a day or two.

Home Emergencies and Power Outages

Most emergencies happen at home: storms, floods, winter outages, wildfires. A little planning makes a huge difference.

Family emergency plan

- Pick two meeting places: one near home, one outside your neighborhood.
- Choose an out-of-area contact person everyone calls or texts. Long-distance lines often work when local ones are jammed.
- Write down important phone numbers on paper. Phones die.
- Know how to turn off your water, gas and electricity.
- Plan for pets, older family members and anyone with medical needs.

Home supplies (at least 3 days, ideally 2 weeks)

- **Water:** about 4 liters (1 gallon) per person per day. See Storing Water at Home.

- **Food** that needs no cooking or little water. See Emergency Food Storage.
- Flashlights, headlamps and spare batteries (avoid candles if you can)
- A battery or hand-crank radio
- First aid kit and at least a week of any prescription medicines
- Phone power banks
- Warm clothing, blankets or sleeping bags
- Cash in small bills
- Copies of important documents in a waterproof bag

During a power outage

- Keep the fridge and freezer **closed**. A closed fridge stays cold for about 4 hours; a full freezer about 48 hours (24 if half full).
- Unplug electronics to protect them from a surge when power returns. Leave one light switched on so you know when it is back.
- In cold weather, close off unused rooms, gather in one room, cover windows with blankets and wear layers.
- In hot weather, stay in the lowest, shadiest room and drink plenty of water.

Generators, grills and camp stoves must never be used indoors, including in garages, basements and on porches. Run generators at least 6 m (20 ft) from any window, door or vent. Fit a battery carbon monoxide alarm.

Food safety after an outage

Throw out perishable food (meat, fish, dairy, eggs, leftovers) that has been above 4 C (40 F) for more than 2 hours. **When in doubt, throw it out.** Never taste food to check it.

If told to evacuate

Go early. Take your emergency kit, medicines, documents, phone chargers and pets. Follow the routes officials give; shortcuts may be blocked or dangerous.

Water

Finding water and making it safe to drink

Finding Water

Where to look

- **Follow the land downhill.** Water collects in valleys, gullies and low ground.
- **Green vegetation** such as willows, cottonwoods, reeds and cattails usually means water is near or just under the surface.
- **Animal trails** often lead to water, especially where several trails join and head downhill.
- **Birds** circling or flying low and straight at dawn and dusk are often heading to water.
- **Insects:** bees and flies usually stay within a few kilometers of water.
- **Dry stream beds:** dig on the outside of bends, where the last water would sit. If the sand is damp, dig a hole and wait for water to seep in.

Collecting water

- **Rain:** spread a tarp, poncho or plastic sheet and funnel it into a container. Rainwater is the cleanest natural source.
- **Dew:** at dawn, wipe grass with a cloth and wring it out.
- **Snow and ice:** melt it first. Eating snow lowers your body temperature. Ice gives more water per pot than snow.
- **Transpiration bag:** tie a clear plastic bag over a leafy, non-poisonous branch in the sun. Water condenses inside.

> Treat all water from the ground, streams, lakes, snow and dew before drinking. Even clear mountain water can carry parasites.

Water to avoid

- Seawater and urine: they dehydrate you faster.
- Water near mines, factories, farm runoff, or with dead animals nearby.
- Water with oily film, strong smell or no living plants around it.

Making Water Safe to Drink

Clear water can still make you very sick. Use one of these methods. If the water is cloudy, let it settle and pour it through cloth first.

Boiling (most reliable)

- Bring water to a **rolling boil for 1 minute**.
- Above 2,000 m (6,500 ft), boil for **3 minutes**.
- Let it cool covered. Boiling kills germs but does not remove chemicals.

Household bleach

Use plain, unscented bleach (no added cleaners or scent). Check the strength on the label.

- **6% bleach:** 8 drops per 4 liters (1 gallon), 2 drops per liter
- **8.25% bleach:** 6 drops per 4 liters (1 gallon), 2 drops per liter
- If the water is cloudy or very cold, **double** the dose.
- Stir and wait **30 minutes**. It should smell slightly of chlorine. If not, repeat the dose and wait 15 more minutes.

Tablets and drops

Chlorine dioxide, iodine or chlorine tablets work well. Follow the package exactly, especially the waiting time. Chlorine dioxide also kills tough parasites like Cryptosporidium if you wait the full time (often 4 hours).

> Iodine is not suitable for pregnant women, people with thyroid problems, or for use over many weeks.

Filters

A good pump or squeeze filter removes bacteria and parasites. Most do not stop viruses. In doubt, filter and then disinfect.

Sunlight (SODIS)

- Fill clear plastic (PET) bottles, 2 liters or smaller, with clear water.
- Lay them in full sun for **6 hours**, or **2 days** if it is cloudy.
- Does not work with cloudy water or in rain.

See also: Improvised Gravity Filter

Improvised Gravity Filter

A layered filter removes mud, debris and some taste. It does **not** make water safe on its own. Always boil or disinfect after filtering.



Layers of an improvised filter

What you need

- A container with a small hole in the bottom: plastic bottle with the bottom cut off, a bucket, or a cloth tied into a cone
- Cloth or grass
- Fine sand
- Charcoal from your fire (not ash), crushed into small pieces
- Small gravel or pebbles

Building it

1. If using a bottle, cut off the bottom and turn it upside down. Plug the neck loosely with cloth.
2. Add layers from the bottom up: cloth, small gravel, fine sand, crushed charcoal, fine sand, then cloth or grass on top.
3. Make each layer several centimeters thick. More sand and charcoal means clearer water.
4. Pour water in slowly. Throw away the first batch; it will carry dust from the charcoal.
5. Run cloudy water through two or three times if needed.

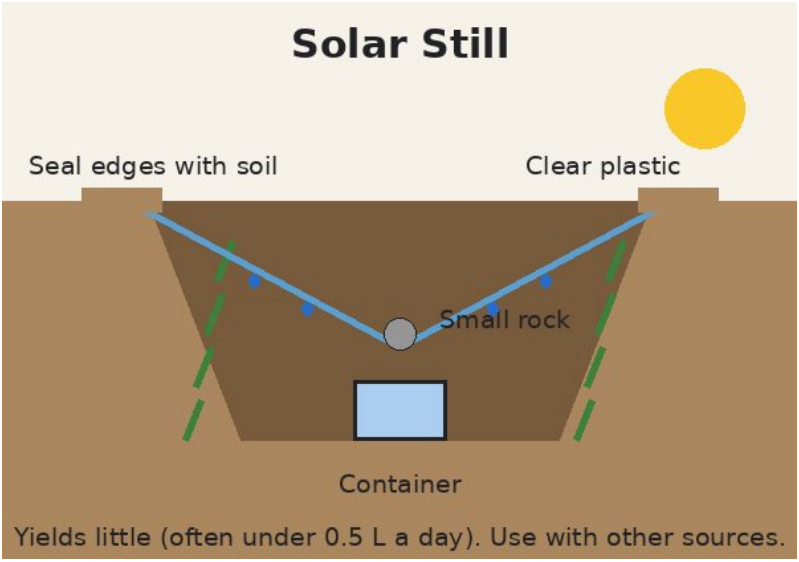
> Charcoal from a campfire is not the same as activated charcoal. It helps with taste and color, but it does not remove chemicals or germs reliably.

After filtering

Boil, use bleach, or use tablets. See Making Water Safe to Drink.

Solar Still

A solar still uses the sun to draw moisture from soil and plants. It makes small amounts, but the water is distilled and clean.



Cross-section of a solar still

Building it

1. Dig a bowl-shaped hole about 90 cm (3 ft) wide and 60 cm (2 ft) deep, in a sunny spot with damp soil if possible.
2. Put a container in the center at the bottom.
3. Add green, non-poisonous plants around the container (not in it) to add moisture.
4. Cover the hole with clear plastic. Seal the edges with soil and rocks so no air escapes.
5. Put a small rock in the middle of the plastic so it forms a cone right above the container.
6. Water condenses on the plastic, runs down to the low point and drips into the container.

Tips

- Use a drinking tube from the container to the outside so you do not need to open the still.
- You can pour salt water or urine into the pit (not the container). The still separates clean water.

- Build several stills if you rely on them.

> A still often yields well under a liter per day, and digging makes you sweat. In hot deserts, do the work at dawn or dusk, and only if there is no better source.

Dehydration and Water Needs

How much you need

- Resting in mild weather: about **2-3 liters a day**.
- Heat or hard work: **4-6 liters or more**.
- Cold and high altitude also dry you out faster than you expect.

> Do not ration water you already have. Drink what you need and work to find more. People have died of dehydration with water still in their bottles.

Signs of dehydration

- **Early:** thirst, dark yellow urine, dry mouth, headache, tiredness.
- **Worse:** little or no urine, dizziness, fast heartbeat, confusion.
- **Severe:** fainting, no sweat in heat, sunken eyes. This is an emergency.

Check your urine. Pale yellow means you are getting enough.

Save water inside your body

- Rest in the shade during the hottest hours. Travel at night or early morning.
- Keep clothes on in the sun; they slow sweating.
- Breathe through your nose and talk less.
- Avoid alcohol, caffeine and salty or very high-protein food when water is short.
- If water is very limited, eat little. Digestion uses water.

Rehydrating

- Sip steadily rather than gulping large amounts.
- If someone has been sweating heavily, add a pinch of salt and a little sugar to a liter of water. Commercial oral rehydration salts are best.

Storing Water at Home

How much

- At least **4 liters (1 gallon) per person per day**: about half for drinking, half for cooking and washing.
- Store at least **3 days' worth**. **Two weeks** is better for home.
- Add more for hot climates, children, nursing mothers, sick people and pets.

Example: a family of 4 for 2 weeks needs about 56 gallons (210 liters).

Containers

- **Best:** sealed, store-bought bottled water. Keep it until the expiry date.
- Food-grade water containers (look for "food grade" or the recycle symbol 1, 2 or 4).
- Clean 2-liter soda bottles work well.
- **Do not use** milk or juice jugs (they hold bacteria and break down), or containers that held chemicals.

Filling your own containers

1. Wash the container with soap and hot water, and rinse.
2. Sanitize: swish a solution of 1 teaspoon of unscented bleach in 1 liter (1 quart) of water around the inside, then empty it.
3. Fill with clean tap water. If your water is not chlorinated (for example, a well), add 2 drops of plain unscented bleach per liter.
4. Cap tightly without touching the inside of the cap.
5. Label it with the date.

Keeping it good

- Store in a cool, dark place, away from fuels and chemicals (plastic lets fumes through).
- Replace home-filled water **every 6 months**.

Hidden water in your home

In an emergency you can also use:

- The water heater tank (turn off the power or gas first, open the drain at the bottom)
- Water in pipes: open the highest tap in the house, then collect from the lowest
- Ice cubes
- The toilet **tank** (not the bowl), only if no chemicals or cleaning tablets are in it, and treat it first

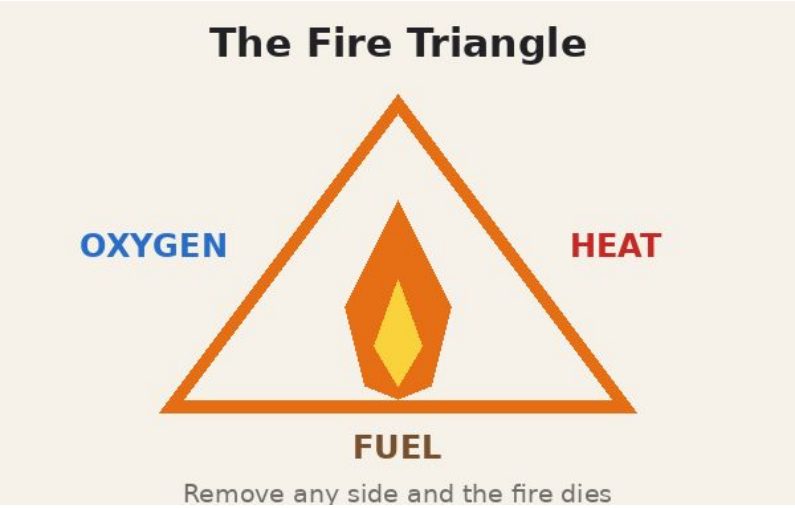
> Treat any of these before drinking. See Making Water Safe to Drink. Never drink from waterbeds, pools or radiators.

Fire

Starting, building and using fire safely

Fire Basics

Fire gives warmth, light, safe water, cooked food, a signal and a big morale boost. Make it before you need it badly.



The fire triangle

What a fire needs

- **Heat** to start it.
- **Fuel** to burn.
- **Oxygen.** Do not pack wood too tightly. A fire that is smothered goes out.

Three sizes of fuel



Tinder, kindling and fuel

- **Tinder** catches a spark or flame: dry grass, birch bark, cedar bark shredded fine, pine needles, cattail fluff, cotton, paper, wood shavings.
- **Kindling** carries the flame: dry twigs from match to pencil thickness.
- **Fuel** keeps it going: sticks and logs from thumb to wrist thickness, then larger.

> Gather all three sizes **before** you strike a light. A new fire dies in the time it takes to go looking for more sticks.

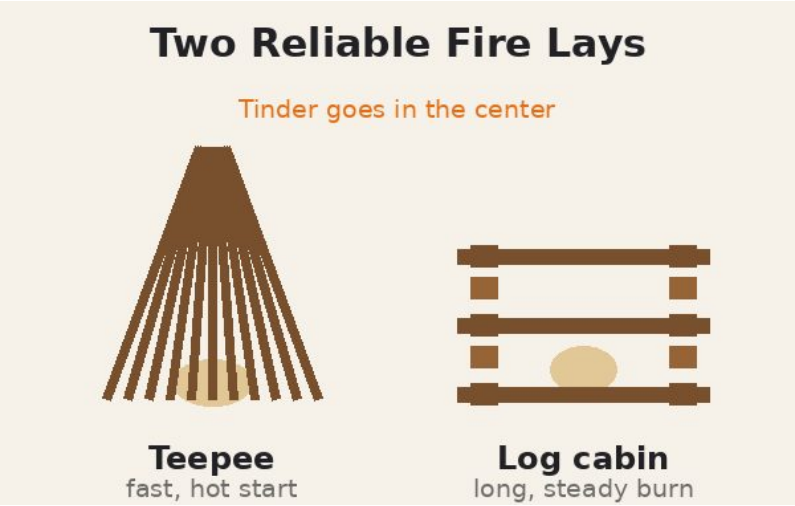
Finding dry wood in wet weather

- Snap test: dry wood breaks with a clean crack. If it bends, it is too wet.
- Take dead branches still on standing trees, not from the ground.
- Look under thick evergreens, inside rotten stumps and under overhangs.
- Split wet sticks; the center is often dry.
- Shave thin curls (feather sticks) from dry wood for kindling.
- Resin-rich pine wood (fatwood) burns even when damp.

Building up

Light the tinder, then add kindling a few pieces at a time. Add larger fuel only when the smaller sizes are burning strongly. Blow gently at the base if it struggles.

Fire Lays



Teepee and log cabin fire lays

Teepee

Put tinder in the center and lean kindling around it in a cone. Leave a gap on the windward side to light it. Burns hot and fast; good for starting and for boiling water.

Log cabin

Start with a small teepee, then stack pairs of sticks around it, each layer at right angles, like a cabin. Gives steady heat and a good bed of coals for cooking.

Lean-to fire

Push a green stick into the ground at an angle over the tinder, or lay one thick log on the windward side. Lean kindling against it. The log shelters the start from wind.

Star fire

Lay several logs like the spokes of a wheel with their ends meeting in the fire. Push them in as they burn. Saves fuel and lasts all night.

Dakota fire hole

Dig a hole about 30 cm (1 ft) deep and wide for the fire. Dig a second hole beside it, on the windward side, and join them with a tunnel at the bottom for air. Burns hot, uses little wood, is hard to see and handles wind well.

> In snow, build the fire on a platform of green logs or rocks so it does not melt into the snow and drown.

Ways to Start a Fire

Carry at least two ways to make fire. Friction methods are a last resort, not a plan.

Lighter and matches

- A cheap disposable lighter is the most reliable tool. Keep one in a warm pocket; cold lighters fail.
- Waterproof your matches in a sealed container. Strike away from your body and shield the flame with your hands.

Ferro rod (fire steel)

1. Place the rod tip right in the tinder.
2. Hold the striker still and pull the rod back hard, so sparks fall into the tinder without knocking it apart.
3. Use very fine, dry tinder: cotton with petroleum jelly, cattail fluff, fine bark shavings.

Battery and steel wool

Touch both terminals of a 9-volt battery to fine steel wool, or bridge a flashlight battery with a strand. The wool glows and lights tinder. Keep steel wool away from batteries when packing.

Lens

A magnifying glass, eyeglasses for far-sightedness, or a clear bottle of water can focus sunlight onto dark tinder. Needs bright sun.

Bow drill (last resort)

1. **Fireboard:** dry, soft, dead wood (willow, cedar, poplar, basswood) about 2 cm thick.
2. **Spindle:** straight, dry stick of the same wood, thumb-thick, about 20 cm long.
3. **Bearing block:** a hard piece of wood or stone with a dent, to press down on the spindle.
4. **Bow:** a curved stick about arm's length with cord tied loosely so you can twist the spindle into it.
5. Cut a notch from the edge of the fireboard to a small burned-in hole. Spin the spindle fast with long, steady bow strokes while pressing down.
6. When smoking dust fills the notch and keeps smoking on its own, you have an ember. Tip it into a tinder bundle and blow gently until it flames.

> Friction fire takes practice and a lot of energy. Learn it before you need it.

Fire Safety

Choosing a site

- Clear a 3 m (10 ft) circle down to bare soil. Remove leaves and dry grass.
- Stay away from overhanging branches, tree roots, peat and dry grass.
- Do not build against a large rock you want to sit by; wet rocks from rivers can crack or burst when heated.
- Pay attention to wind. Sparks carry far in dry weather.

Carbon monoxide

> Never burn a fire, stove, charcoal or generator inside a closed tent, car, snow cave or small room. Carbon monoxide has no smell and kills in your sleep.

Signs of poisoning: headache, dizziness, nausea, confusion, sleepiness. Get the person into fresh air at once.

Putting a fire out

1. Let it burn down to ash if you can.
2. Pour water on it and stir the ashes. Pour more and stir again.

3. If water is scarce, smother with soil or sand and stir, but do not just bury it; buried coals can stay hot for a day.
4. Feel for heat with the back of your hand held close. If it is too hot to touch, it is too hot to leave.

If a wildfire threatens

- Move to areas with little fuel: rock, gravel, a recently burned area, or across a wide river.
- Move downhill and across the slope, not uphill. Fire races uphill.
- Cover your skin with wool or cotton, not synthetics, and breathe through a damp cloth.

Cooking Over a Fire

Cook on coals, not flames

Flames burn the outside and leave the inside raw. Let a fire burn down to a bed of glowing coals, or rake coals to one side of the fire to cook on while the other side keeps burning.

Hand test for heat: hold your palm about 10 cm (4 in) above the coals.

- 2 seconds before it is too hot: very hot
- 4-5 seconds: medium, good for most cooking
- 6-8 seconds: low, good for simmering

Methods

- **Boiling:** the safest way to cook anything you are unsure about, and it makes water safe. Hang the pot from a tripod or support it on rocks or green logs.
- **Foil packets:** wrap food in two layers of foil and set it on coals. Turn once. Vegetables take 15-20 minutes, fish about 10.
- **Skewers and spits:** use green (live) wood, so it does not burn through. Avoid poisonous woods like yew and oleander.
- **Hot stone:** heat a flat, dry stone at the edge of the fire and cook on it like a pan.

- **Pit cooking:** heat stones in a fire in a pit, rake out the fire, wrap food in green leaves, cover with soil and leave for hours.

> Never heat rocks from rivers, lakes or wet ground. Trapped water can make them crack or explode.

Cook meat fully

Wild meat and fish can carry parasites. Cook until juices run clear and there is no pink.

- Poultry and birds: 74 C (165 F) inside
- Ground or chopped meat: 71 C (160 F)
- Whole cuts of game: 71 C (160 F) is safest for wild meat
- Fish: until it flakes easily and looks opaque

Safety

- Keep a pot of water or a pile of soil next to the cooking fire.
- Use a stick or cloth to handle hot pots; metal handles get very hot.
- Wash hands before handling food, and keep raw meat away from cooked food.

Shelter

Staying warm, dry and out of the weather

Shelter Basics and Site Choice

In cold, wet or windy weather, shelter is your top priority after first aid. A good shelter is small, dry, out of the wind and insulated from the ground.

How you lose heat

- **Conduction:** touching cold ground or rock. The ground steals heat fastest, so insulate under you first.
- **Convection:** wind blowing heat away.
- **Evaporation:** wet clothes and sweat.
- **Radiation:** heat leaving your body to the open sky.

Choosing a site

- Flat, slightly raised and well-drained ground.
- Near building materials, firewood and water, but at least 60 m (200 ft) from streams.
- Out of the wind: behind rocks, thick trees or a ridge.

Avoid

- **Dead trees and hanging branches** ("widowmakers") above you.
- Dry stream beds and valley floors (floods, and cold air sinks there at night).
- Bases of cliffs (rockfall) and avalanche slopes.
- Insect nests and animal trails.

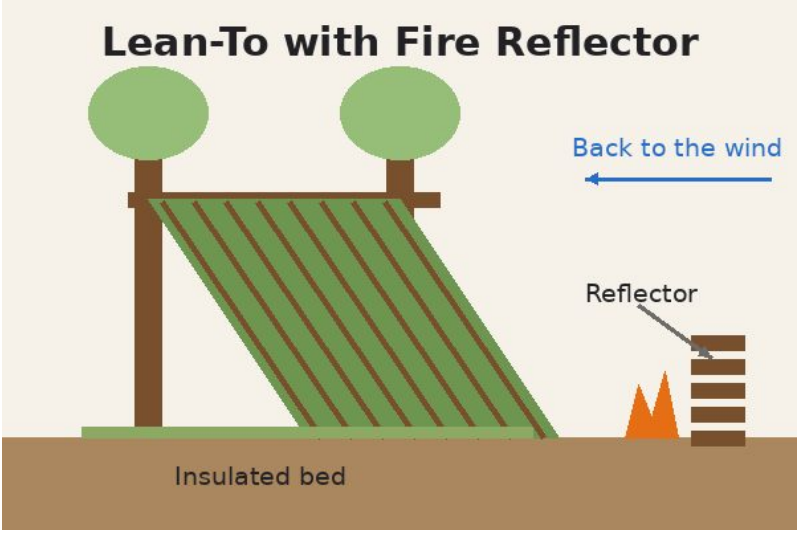
> Pile at least 10-20 cm (4-8 in) of dry leaves, grass, pine boughs or a pack between you and the ground, even inside a tent. It compresses when you lie on it.

Before dark

Start your shelter at least two hours before sunset. Building in the dark is slow and dangerous.

Lean-To Shelter

A lean-to is quick to build and pairs well with a fire in front. It blocks wind and rain from one side.



Lean-to with fire reflector

Building it

1. Find two trees about 2-3 m apart, or drive two forked sticks into the ground.
2. Tie or wedge a strong **ridge pole** across them at about chest height.
3. Lean long poles against the ridge pole on the windward side, close together, at about 45-60 degrees.
4. Weave smaller branches across the poles.
5. Cover with layers of bark, leafy branches, grass or a tarp, starting at the bottom so each layer overlaps the one below, like shingles.
6. Make a thick bed of dry insulation inside.

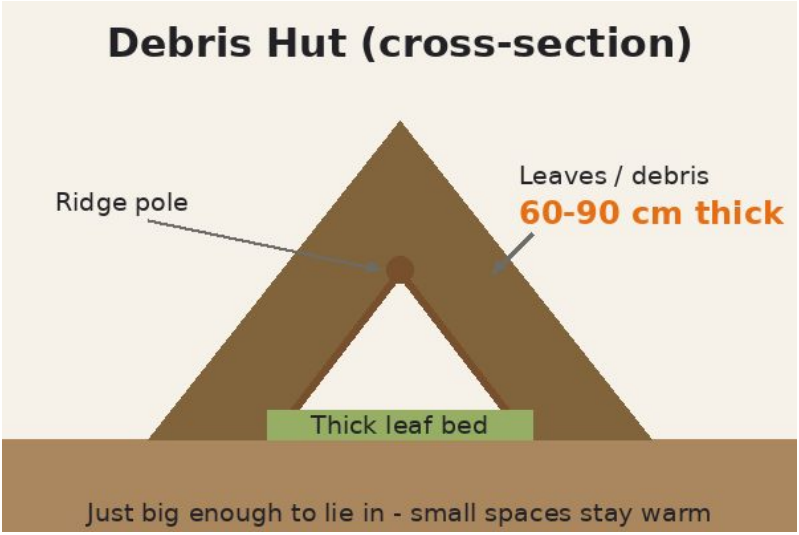
Adding a fire reflector

Build a fire about one step in front of the open side. Behind the fire, stack green logs between stakes to make a wall. It throws heat back into the shelter.

> The back of the lean-to must face the wind. Check the wind direction before you start, and watch for changes.

Debris Hut

The debris hut keeps you warm with no fire, sleeping bag or tools. It works like a giant sleeping bag made of leaves.



Debris hut cross-section

Building it

1. Find a **ridge pole** a bit longer than your height.
2. Prop one end on a stump, rock or a pair of crossed sticks about waist high. The other end rests on the ground.
3. Lie down under it to check: it should be just big enough to fit you. Bigger means colder.
4. Lean sticks along both sides of the pole like ribs, close together.
5. Lay a lattice of small branches over the ribs so leaves do not fall through.
6. Pile on dry leaves, grass, pine needles and debris until the walls are **60-90 cm (2-3 ft) thick**. Thicker for colder nights.
7. Add some branches on top to keep the wind from blowing the leaves off.

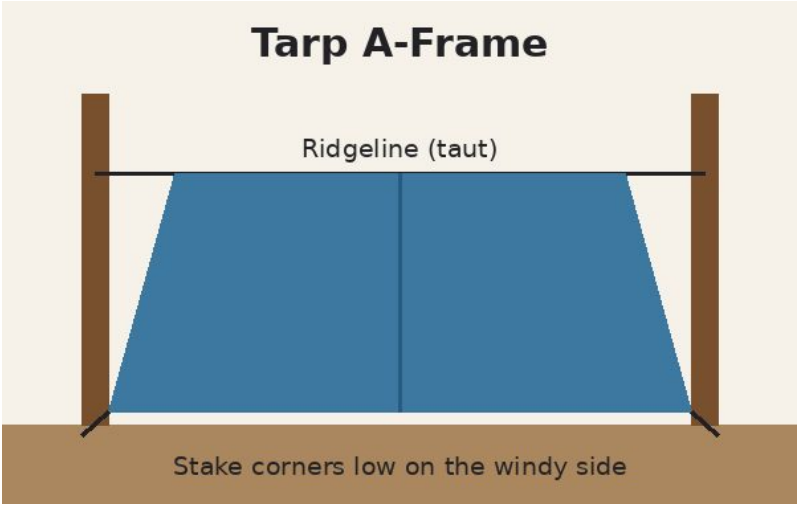
8 Stuff the inside with dry leaves for a thick bed, and keep a pile by the entrance.

9 Crawl in feet first and pull the leaf pile in to plug the door.

> It takes a lot of debris, often a few hours of work. Start early and gather more than you think you need.

Tarp and Poncho Shelters

A tarp is the fastest shelter there is. Even a large trash bag or emergency blanket can become a roof.



Tarp A-frame

A-frame

1 Tie a cord tight between two trees at waist to chest height.

2 Hang the tarp over the cord so both sides are equal.

3 Stake or weight the corners down. Pull them out until the tarp is tight.

4 In wind, set the windward side lower or right to the ground.

Lean-to tarp

Tie the top edge of the tarp along the ridgeline and stake the bottom edge to the ground on the windward side. Good with a fire in front.

Tips

- No grommets? Wrap a small stone in the tarp corner and tie the cord around it.
- Angle the roof steeply so rain runs off.
- Dig a shallow trench uphill of the shelter to steer water around it.
- Tie cord with a knot you can untie: a taut-line hitch or truckers hitch lets you tighten the line.

> An emergency blanket reflects heat back to you, but it holds condensation. Wrap it outside your insulation, not against your skin, when you can.

Cold Weather and Snow Shelters

Stay warm with COLD

- **C - Clean:** dirty clothes lose insulation.
- **O - avoid Overheating:** remove layers before you sweat.
- **L - Loose layers:** tight clothing cuts circulation.
- **D - Dry:** wet clothing pulls heat away fast. Dry socks matter most.

Snow trench

1. Dig a trench as long as your body, about 1 m (3 ft) deep, in firm snow.
2. Roof it with branches, skis or a tarp, then pile snow on top.
3. Line the floor with thick insulation: boughs, packs, foam.

Quinzee (snow mound)

1. Pile snow into a mound about 2 m (7 ft) high. Mix layers as you pile.
2. Let it settle for 1-2 hours.
3. Push sticks about 30 cm (1 ft) into the mound all over, to show wall thickness.
4. Dig a low entrance and hollow it out until you hit the sticks.

5 Poke a fist-sized **air hole** in the roof.

> Always make an air vent in any snow shelter and keep it clear. Keep a digging tool inside with you.

Tree well

Under a large evergreen, snow is often shallow near the trunk. Dig out the space and roof it with boughs.

Body heat tips

- Eat before sleeping; digestion makes heat.
- Do light exercise (not to the point of sweating) before getting in your shelter.
- Put a bottle of hot water inside your clothes or bag.
- Share body heat with others.

Desert and Hot Weather Shelter

In heat, shelter means **shade**. The goal is to save water by not sweating.

Simple shade

- Use natural shade: rock overhangs, the shady side of boulders, bushes.
- Hang a tarp or blanket 30-45 cm (12-18 in) above the ground, with a second layer above it if you have it. The air gap between layers cuts heat a lot.

Below-ground shelter

1 Dig a trench 45-60 cm (18-24 in) deep, as long as your body, oriented east-west.

2 Pile the sand around the edges.

3 Cover with a tarp, then a second layer with a gap between them.

The ground 30 cm down can be much cooler than the surface. Dig at dawn or dusk, not midday.

Habits

- Rest in the heat of the day; work and travel at night or early morning.

- Keep your skin covered with loose, light-colored clothing and a hat.
- Do not lie directly on hot ground. Sit on a pack or brush.
- Watch for snakes and scorpions in shade spots; check before you reach in.

> Cars, metal sheds and tents in full sun can become deadly hot. Shade with airflow is better.

Sleeping Warm

A cold night is miserable and dangerous, and it wrecks your judgment the next day. These tricks work with or without good gear.

Before bed

- **Eat** something, especially fats and sugars. Your body burns food to make heat. Keep a snack by your bed for the middle of the night.
- **Drink** warm water. Being dehydrated makes you colder.
- **Pee** before you lie down. Your body wastes heat keeping a full bladder warm.
- **Warm up** with light exercise (not to the point of sweating) right before getting in.
- **Change into dry clothes**, especially socks. Put damp clothes in your bag to dry them with body heat if you have no other way.

Under you matters most

The ground pulls heat from you faster than the air. Put down **at least 10-20 cm (4-8 in)** of dry insulation: leaves, grass, pine boughs, a foam pad, an empty pack, a coiled rope. Then more.

Around you

- Wear a **hat** and cover your neck.
- Fill air gaps in your sleeping bag or blankets with spare clothes.
- Fill a bottle with hot water (check the lid seals and the bottle can take heat), wrap it in a sock, and put it between your thighs or on your stomach.
- Keep your face out of the bag; breath adds moisture that makes it colder.

- Huddle together with others if you can.

Through the night

- If you wake up cold, eat, do some sit-ups or clench and relax your muscles, then try again.
- Keep tomorrow's water and boots inside your shelter or bag so they do not freeze.

> Shivering is a warning. If someone shivers hard and cannot stop, or starts stumbling and mumbling, treat for Hypothermia now.

First Aid

Emergency care when help is far away

First Aid Priorities

> This wiki is a field reference, not medical training. Take a first aid or wilderness first aid course. Get professional care as soon as you can.

1. Make the scene safe

Check for traffic, fire, falling rocks, water, electricity or an animal. Do not become a second casualty.

2. Check the person

Tap their shoulder and shout. Are they awake and answering?

3. Call or send for help

If you have any signal, call your local emergency number (911 in North America, 112 in Europe and many countries). Text messages sometimes get through when calls do not.

4. Work through the big killers, in order

- **Massive bleeding:** stop it first. See Severe Bleeding.
- **Airway:** open it with head tilt and chin lift. Clear anything visible in the mouth.
- **Breathing:** look, listen and feel for up to 10 seconds. Gasping is not normal breathing.
- **No normal breathing:** start CPR.
- **Circulation:** look for other bleeding and signs of Shock.

5. Then check head to toe

Look and gently feel for injuries: head, neck, chest, belly, pelvis, arms and legs. Ask what hurts. Note what you find and the time.

Recovery position

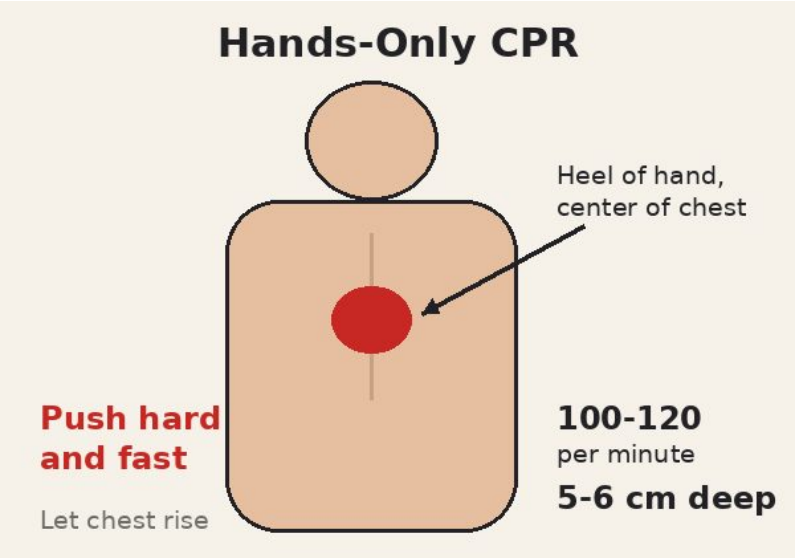
For someone breathing but not awake (and with no suspected spine injury): roll them onto their side, top knee bent to hold them there, head tilted back slightly so fluids drain from the mouth. Check breathing often.

Protect them

Keep them warm, off the cold ground, and calm. Do not give food or drink to someone who is badly injured or not fully alert.

CPR

Start CPR if a person is **unresponsive and not breathing normally** (or only gasping).



Hands-only CPR

Adults

1. Call for help. If someone else is there, send them to call and to find an AED (defibrillator).
2. Lay the person on their back on a firm surface.
3. Put the heel of one hand on the **center of the chest** (lower half of the breastbone), the other hand on top.
4. Arms straight, shoulders over your hands. Push **hard and fast**:

- **Depth:** 5-6 cm (2-2.4 in)
- **Rate:** 100-120 per minute (the beat of the song "Stayin' Alive")

- Let the chest come all the way back up between pushes.

1. If trained and willing: after every 30 compressions, give 2 rescue breaths (head tilt, chin lift, pinch nose, breathe until the chest rises).
2. If not trained: keep doing compressions only. That still saves lives.
3. Keep going until the person starts breathing, an AED is ready, help takes over, or you are too exhausted to continue.

AED

Turn it on and follow the voice instructions. Keep compressions going while it is being set up.

Children and infants

- **Child:** one or two hands, about 5 cm (2 in) deep.
- **Infant:** two fingers in the center of the chest just below the nipple line, about 4 cm (1.5 in) deep.
- For children and drowning victims, rescue breaths are especially important if you can give them.

> In remote places with no chance of advanced care, CPR rarely restarts the heart after a long time. It is still right to try. Your safety and the group's safety come first.

Severe Bleeding

Heavy bleeding can kill in minutes. Act fast.

1. Direct pressure

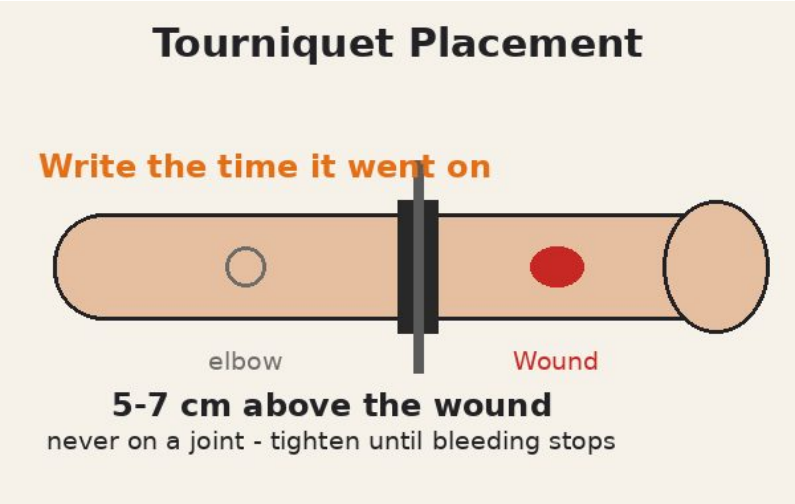
- Put a cloth or gauze on the wound (bare hands if nothing else) and **press hard** with both hands.
- Hold pressure for at least **10 minutes** without lifting to check.
- If blood soaks through, add more material on top and keep pressing.

2. Pack deep wounds

For a deep wound in the thigh, armpit, groin or neck, push gauze or clean cloth **into** the wound, filling it, then press hard on top for at least 3 minutes (10 if plain cloth).

3. Tourniquet for arms and legs

Use a tourniquet when bleeding from an arm or leg is life-threatening and pressure does not stop it, or you cannot hold pressure (for example, several injured people).



Tourniquet placement

1. Place it **5-7 cm (2-3 in) above the wound**, never on a joint. If unsure, place it high on the limb.
2. Tighten until the bleeding **stops**. It will hurt a lot. That is expected.
3. **Write the time** on the tourniquet or the person's skin.
4. Do not loosen it. Leave it for medical staff.
5. If bleeding continues, add a second tourniquet just above the first.

Improvised tourniquet

A strip of cloth at least 4 cm (1.5 in) wide, tied around the limb, with a stick through the knot. Twist the stick until bleeding stops, then tie the stick in place. Thin cord or wire cuts the skin and does not work well.

> After bleeding is controlled, keep the person warm and lying down. Watch for Shock.

Choking

If the person can cough or speak, encourage them to keep coughing. Act when they **cannot breathe, cough or speak**.

Adults and children over 1 year

1. Stand behind them. Lean them forward.
2. Give up to **5 firm back blows** between the shoulder blades with the heel of your hand.
3. If that does not work, give up to **5 abdominal thrusts**: make a fist just above the belly button, grab it with your other hand, and pull sharply **in and up**.
4. Keep alternating 5 back blows and 5 thrusts until the object comes out.
5. Use chest thrusts instead of abdominal thrusts for pregnant women and very large people.

Infants (under 1 year)

1. Lay the baby face down along your forearm, head lower than the body, supporting the jaw.
2. Give 5 back blows between the shoulder blades.
3. Turn the baby face up and give 5 chest thrusts with two fingers in the center of the chest.
4. Repeat. Never do abdominal thrusts on an infant.

If they become unresponsive

Lower them to the ground and start CPR. Look in the mouth before rescue breaths and remove any object you can see. Do not sweep blindly with your finger.

> Anyone who received abdominal thrusts should be checked by a medical professional when possible.

Shock

Shock happens when the body cannot get enough blood to the organs, often after bleeding, burns, severe pain or dehydration. It can kill even after the injury

seems handled.

Signs

- Pale, cool, clammy skin
- Fast, weak pulse and fast breathing
- Anxiety, restlessness or confusion
- Thirst, nausea
- Later: drowsiness and unresponsiveness

What to do

1. Stop any bleeding.
2. Lay the person down on their back.
3. If there is no injury to the head, neck, back, hips or legs, you may raise the legs about 30 cm (12 in).
4. **Keep them warm.** Put insulation under them and cover them. Cold makes shock worse.
5. Keep them calm and reassure them.
6. Do not give food or drink if they may need surgery soon. In a remote setting where help is many hours away and the person is fully alert, small sips of water are reasonable.
7. Check breathing and pulse often. Start CPR if they stop breathing normally.

> If vomiting or not fully awake, turn them onto their side to protect the airway.

Hypothermia and Frostbite

Hypothermia can happen well above freezing when someone is wet, tired and in the wind.

Signs - watch for the "umbles"

- **Mild:** shivering, stumbling, fumbling hands, mumbling, poor decisions.
- **Moderate to severe:** shivering slows or stops, confusion, drowsiness, slurred speech, acting strangely (some people even undress).

Treatment

1. Get them out of the wind and wet. Shelter first.
2. Replace wet clothes with dry ones, or cover wet clothes with a waterproof layer if you cannot change them.
3. Insulate from the ground, and wrap the whole body, including the head and neck.
4. Put warm (not hot) water bottles or heat packs, wrapped in cloth, on the chest, armpits and back.
5. If fully awake and able to swallow: give warm, sweet drinks and food.
6. Share body heat in a sleeping bag if needed.

> Handle a severely cold person very **gently** and keep them lying flat. Rough movement can trigger a fatal heart rhythm. Do not rub the skin. No alcohol.

If they are unresponsive and not breathing normally after checking for up to a full minute, start CPR.

Frostbite

- **Signs:** numb, white or grey-yellow skin that feels hard or waxy. Fingers, toes, nose, ears.
- Do not rub it, and do not use direct heat like a fire or stove.
- **Only rewarm if it will not freeze again.** Refreezing does much more damage.
- Rewarm in warm water about 37-39 C (99-102 F), comfortable to your own hand, for 15-30 minutes. It will be very painful.
- Do not pop blisters. Cover loosely with dry, clean dressing and put gauze between fingers and toes.

Heat Exhaustion and Heat Stroke

Heat exhaustion

Signs: heavy sweating, pale or flushed skin, weakness, dizziness, headache, nausea, cramps, fast pulse.

1. Move to shade and rest.
2. Loosen and remove extra clothing.
3. Cool with wet cloths and fanning.
4. Give water or a salt-and-sugar drink in small sips.
5. Rest for the rest of the day. If not better within an hour, treat as heat stroke.

Heat stroke (life-threatening)

Signs: very hot body, **confusion, odd behavior, slurred speech, seizure or collapse**. Skin may be dry or still sweating.

> Any confusion in a hot, overheated person is heat stroke until proven otherwise. Cool first, transport second.

1. Cool them **immediately and aggressively**.
2. Best: put them in cool or cold water up to the neck (a stream, lake or tub), keeping the head above water and staying with them.
3. If you cannot: soak them with water and fan hard. Put cold packs or wet cloths on the neck, armpits and groin. Keep re-wetting.
4. Keep cooling until they are thinking clearly again.
5. Do not give drinks to someone who is confused or not fully awake.
6. Get medical help. Heat stroke can damage organs even after recovery.

Prevention

Drink regularly, rest in the shade, work in the cool hours, and watch each other for early signs.

Burns

Stop the burning

- If clothing is on fire: stop, drop and roll, or smother with a blanket.
- Remove rings, watches and tight clothing near the burn before it swells. Do not pull off clothing stuck to the skin.

Cool it

- Cool with clean, cool running water for **20 minutes** (at least 10). This helps even if started late, up to about 3 hours after.
- Do **not** use ice, butter, oil, toothpaste or ointment on a fresh burn.
- Keep the rest of the person warm. Cooling a large burn can cause hypothermia, especially in children.

Cover it

- Cover loosely with clean plastic wrap (laid on, not wrapped tight), a clean non-stick dressing, or a clean plastic bag over a hand.
- Do not pop blisters.

Serious burns - get help

- Larger than the person's palm, or deep (white, leathery or charred)
- On the face, hands, feet, genitals or over joints
- Electrical or chemical burns
- Burns in the airway: soot around the nose or mouth, hoarse voice, coughing, trouble breathing. These can swell shut. **Urgent.**

Chemical burns

Brush off dry chemicals, then flush with lots of water for at least 20 minutes.

> Large burns cause dangerous fluid loss. Watch for Shock and give sips of fluid if the person is alert and help is far away.

Broken Bones and Sprains

Signs of a break

Pain, swelling, bruising, deformity, grating, or the person cannot use the limb. If in doubt, treat it as broken.

Splinting

1. Control any bleeding first. Cover open wounds and bone ends with a clean dressing. Do not push bone back in.

2. **Check circulation, feeling and movement** beyond the injury (fingers or toes): warm, pink, can feel your touch, can wiggle.
3. Splint the limb in the position you find it, unless it is badly bent and there is no pulse beyond it (then gently straighten it once, along its length, if trained).
4. Immobilize the **joint above and below** a broken bone, or the bones above and below an injured joint.
5. Use anything rigid: sticks, trekking poles, rolled magazines, a sleeping pad. Pad well with clothing.
6. Tie in place firmly but not tightly, with knots away from the injury.
7. **Re-check circulation** after splinting and every 30 minutes. If fingers or toes go pale, cold or numb, loosen the ties.

Slings

For arm, wrist and collarbone injuries, support the arm in a sling made from a triangle of cloth or a shirt, with the hand slightly higher than the elbow.

Sprains

Rest, cool packs wrapped in cloth for 20 minutes at a time, gentle compression wrap, and raise the limb. A sprained ankle may be walkable with a firm wrap and a stick if you must move.

> Suspect a **spine injury** after a fall, crash or blow to the head with neck or back pain, numbness or tingling. Keep the head and neck still and do not move the person unless they are in danger.

Wound Care and Infection

In the wild, infection is a bigger long-term danger than most cuts.

Cleaning

1. Stop the bleeding with pressure.
2. **Irrigate** with lots of drinkable water: at least 1 liter for a wound. Squirt it with force, for example from a bag with a pinhole or a syringe.
3. Remove dirt and splinters with clean tweezers.

4 Clean the skin around the wound with soap and water.

Closing and covering

- Small, clean cuts can be closed with tape strips.
- Leave dirty wounds, bites and punctures **open**, covered with a clean dressing, so they can drain.
- Change dressings daily, or when wet or dirty.

Signs of infection

- Redness and swelling that spreads
- Increasing pain, warmth and pus
- **Red streaks** moving up from the wound, fever or chills: this is serious. Get to medical care urgently.

If infected: open the wound if it was closed, soak in warm clean water several times a day, keep it clean and covered.

Blisters

Cover hot spots early with tape. For large blisters, drain from the edge with a clean needle, leave the skin on, and pad around it.

> Animal bites can carry rabies. Wash any bite for 15 minutes with soap and water and get medical care as soon as possible.

Snakebite

Most snakebites are not fatal, especially with calm, correct first aid and medical care.

Do

1 Move away from the snake. Do not try to catch or kill it; a photo from a distance is enough.

2 Keep the person calm and as still as possible.

3 Remove rings, watches and tight clothing from the bitten limb before swelling starts.

4. Keep the bitten limb still and at about heart level. Splint it if you can.
5. Mark the edge of swelling with a pen every 30 minutes, with the time.
6. Get to medical care as fast as safely possible. Carrying the person is better than letting them walk, if practical.

Do not

- Cut the wound or try to suck out venom
- Use a tourniquet
- Apply ice or heat
- Give alcohol or medicines like aspirin
- Use electric shock or "snakebite kits"

Regional note

In **Australia**, and for some snakes elsewhere (such as coral snakes, cobras and mambas), a **pressure immobilization bandage** is advised: wrap a wide elastic bandage firmly from the bite up the whole limb, then splint. Do **not** use this for North American rattlesnakes and other pit vipers, where it can increase tissue damage.

> Signs needing urgent help: trouble breathing or swallowing, drooping eyelids, bleeding gums, vomiting, fainting, rapidly spreading swelling.

Severe Allergic Reaction (Anaphylaxis)

Anaphylaxis is a fast, life-threatening allergic reaction, usually to stings, foods, medicines or latex. It can kill within minutes.

Signs (often within minutes of contact)

- Trouble breathing, wheezing, noisy breathing, tight throat, hoarse voice
- Swelling of the lips, tongue or face
- Hives or widespread rash, flushing
- Dizziness, fainting, pale and floppy (especially in children)
- Vomiting, stomach cramps

- A feeling that something terrible is happening

What to do

1. **Use an epinephrine auto-injector right away** if the person has one. Do not wait to see if it gets worse.
2. Inject into the **outer middle of the thigh**. It can go through clothing. Follow the instructions printed on the device and hold it in place for the time it says.
3. **Call for emergency help**, even if they feel better. Symptoms can come back hours later.
4. **Positioning:**

- Lie them down with legs raised.
- If breathing is hard, let them sit up instead.
- If vomiting or unconscious, put them on their side.
- Do not let them stand or walk suddenly, even if they feel better. This can cause collapse.

1. If there is no improvement after **5 minutes** and a second auto-injector is available, give the second one.
2. If they stop breathing normally, start CPR.

> Antihistamine tablets and inhalers do not stop anaphylaxis. Epinephrine is the treatment. Give it first; other medicines can come after.

Remove the cause

- Stings: scrape the stinger out sideways with a fingernail or card. Do it quickly.
- Food: do not make them vomit.

Prevention

Anyone with a known severe allergy should carry **two** auto-injectors at all times, and the people with them should know where they are and how to use them.

Head Injuries and Concussion

After any blow to the head

- If they were knocked out, even briefly, treat it as serious.
- Assume a **neck injury** too if they fell from a height, were in a crash, or have neck pain, numbness or tingling. Keep the head and neck still. See Broken Bones and Sprains.
- Control scalp bleeding with firm pressure. Scalps bleed a lot even from small cuts. Do not press hard if the skull feels soft or dented; press around the edges instead.

Concussion signs

Headache, dizziness, nausea, confusion, blurred vision, ringing ears, memory gaps around the injury, feeling "foggy," trouble concentrating.

Danger signs - get emergency help urgently

- Getting **more drowsy** or hard to wake
- Headache that gets **worse** and will not go away
- **Repeated vomiting**
- Confusion, strange behavior, slurred speech
- **One pupil larger** than the other
- Weakness, numbness or trouble moving on one side
- **Seizure**
- Clear fluid or blood from the nose or ears
- Bruising behind the ears or around both eyes

Caring for someone with a mild head injury

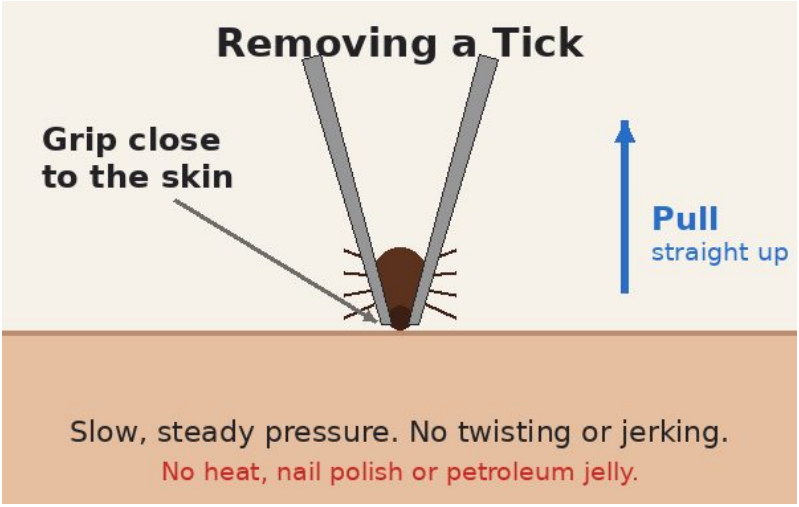
- Rest. No hard physical work, and no more risk of another hit.
- Have someone stay with them for at least the first 24 hours and check on them regularly. If they are asleep and cannot be woken normally, that is an emergency.
- Avoid alcohol.
- If unconscious but breathing, put them in the recovery position (on their side), supporting the head and neck.

Older adults and people on blood thinners can bleed inside the skull after even a minor bump. Signs may take hours or days. Watch them closely.

Ticks, Insect Stings and Spider Bites

Ticks

Ticks can spread Lyme disease and other illnesses. The sooner you remove one, the lower the risk.



Removing a tick

1. Use fine-tipped tweezers. Grab the tick **as close to the skin as possible**.
2. Pull **straight up** with slow, steady pressure. Do not twist or jerk.
3. If mouth parts stay in the skin and you cannot easily remove them, leave them; the skin will heal.
4. Clean the bite and your hands with soap and water or alcohol.
5. Do not use heat, nail polish or petroleum jelly. They make the tick release more fluid into you.

Watch for a spreading rash (often a "bull's-eye" ring), fever, chills, aches or tiredness in the next 30 days. Get medical care if they appear and say you were bitten.

Prevention: long sleeves and trousers tucked into socks, repellent, and a full-body tick check every evening, including scalp, armpits, groin and behind knees.

Bee and wasp stings

1. Scrape out a bee stinger quickly with a fingernail or card edge.
2. Wash the area.
3. Cool with a cold, wet cloth to ease swelling and pain.
4. **Watch for anaphylaxis:** trouble breathing, face or throat swelling, dizziness, hives spreading. See Severe Allergic Reaction.
5. Many stings at once, or a sting inside the mouth or throat, needs urgent help.

Spider bites

Most spider bites are harmless. Clean, cool and watch them.

Get medical care for:

- **Widow spiders** (shiny black, red hourglass mark): severe muscle cramps, stomach pain, sweating.
- **Recluse spiders** (brown, violin shape on the back): a bite that turns into a spreading, dark or open sore over hours or days.
- Any bite with spreading redness, fever or feeling unwell.

Scorpions

Clean the sting and cool it. Most cause sharp pain only. Get urgent help for children, or for anyone with numbness spreading, twitching, roving eye movements, drooling or trouble breathing.

Seizures

During a seizure

1. **Stay calm and time it.**
2. Move hard or sharp objects away. Put something soft under their head.
3. Loosen anything tight around the neck.

4 **Do not hold them down** and **do not put anything in their mouth**. They cannot swallow their tongue.

5 Stay with them until it ends.

After it stops

- Roll them onto their **side** in the recovery position to keep the airway clear.
- Check their breathing.
- They will likely be confused and very tired. Speak calmly, tell them what happened, and let them rest.
- Do not give food or drink until they are fully awake.

Get emergency help if

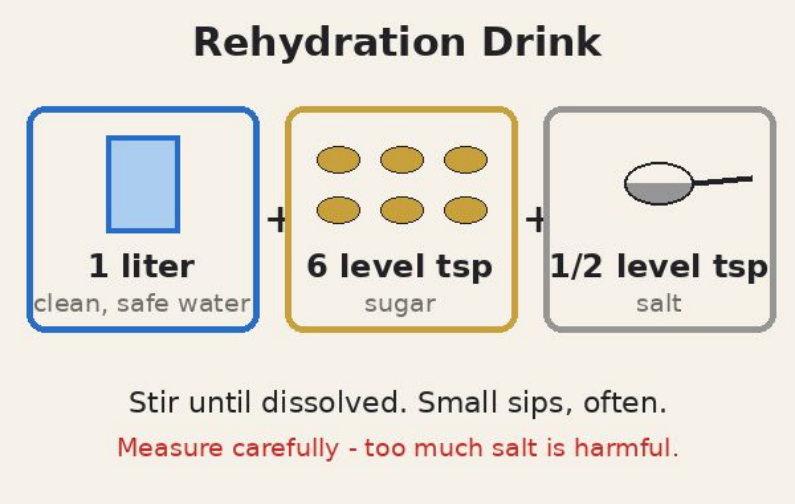
- The seizure lasts **more than 5 minutes**
- Another seizure starts soon after the first
- They do not wake up, or have trouble breathing afterward
- It is their first seizure, or you do not know their history
- It happened in water
- They are injured, pregnant, or have diabetes
- They have a high fever or recent head injury

> Seizures can also be caused by heat stroke, very low blood sugar, head injury or poisoning. Treat the cause if you can: cool someone overheated, and give sugar to a diabetic once they are awake and can swallow.

Diarrhea, Vomiting and Rehydration

In survival settings, diarrhea from bad water or food is common, and dehydration from it can be deadly, especially for children and older people.

Rehydration drink



Homemade oral rehydration solution

- **1 liter** of clean, safe water
- **6 level teaspoons** of sugar
- **1/2 level teaspoon** of salt

Stir until dissolved. Measure carefully: too much salt is harmful. Commercial oral rehydration salts (ORS) are best if you have them.

How to give it

- Small sips often, every few minutes, even if they are vomiting. Most of it will stay down.
- Adults: aim for at least a cup after each loose stool, on top of normal drinking.
- Children: a quarter to half cup after each loose stool, by spoon if needed.
- Keep breastfeeding babies.
- Once vomiting settles, start eating simple foods: rice, porridge, bread, bananas, soup.

Danger signs - get help

- Blood in the stool or black, tarry stool
- High fever

- No urine for 8 hours or more, very dark urine
- Very sleepy, confused, or unable to drink
- Sunken eyes, dry mouth, no tears (in children)
- Diarrhea lasting more than 3 days, or vomiting that will not stop for a day

Stop it spreading

- **Wash hands** with soap after the toilet and before handling food or water.
- Dig toilets at least 60 m (200 ft) from water sources, camp and trails, and 15-20 cm (6-8 in) deep. Cover waste completely.
- Treat all drinking water. See Making Water Safe to Drink.
- Keep a separate cup and towel for the sick person.

Eye Injuries

Dust or grit in the eye

1. Do not rub it.
2. Blink several times and let tears wash it out.
3. Flush with clean water from the inner corner (nose side) outward.
4. Pull the upper lid down over the lower lashes, which can sweep the speck off.
5. If it is on the white of the eye, lift it gently with the corner of a clean, wet cloth.

Chemicals in the eye

Flush **immediately** with lots of clean water for at least **15-20 minutes**, holding the eye open, with the injured eye lower so water does not run into the other eye. Then get medical help.

Object stuck in the eye

> Do not try to remove anything stuck in the eye.

1. Keep the person still and calm.
2. Cover the injured eye with a paper cup or similar shield taped in place, without pressing on the eye.

3 Cover the other eye too. Eyes move together, so this keeps the injured eye still.

4 Get medical help urgently.

Blow to the eye

Cool with a cold cloth for 15 minutes, without pressure. Seek help for vision changes, blood inside the colored part of the eye, severe pain, or if they cannot move the eye normally.

Snow blindness

Sunburn of the eye from sun reflecting off snow, water or sand. Painful, gritty, red, watery eyes, often hours later.

- Rest the eyes in the dark, with cool cloths over them.
- It usually heals in 1-2 days.
- **Prevent it:** wear sunglasses. If you have none, make slit goggles from bark, cardboard or cloth with thin horizontal slits, and darken the skin under your eyes with charcoal.

Food

Finding and preparing food safely

Food in a Survival Situation

Food is the last of the big survival needs. Most healthy people can go weeks without it. Hunger is unpleasant, but it is rarely what kills.

Why food still matters

- It keeps you warm, especially at night in cold weather.
- It keeps your mind sharp and your mood up.
- Long-term survival, past a few days, needs a food plan.

Rules of thumb

- **Do not spend more energy finding food than it gives you.** Chasing animals is usually a losing trade.
- Eat high-energy foods you carried first, in small regular amounts.
- If water is very short, eat little. Digestion, especially of protein, uses water.
- Easy calories: insects, fish, shellfish from clean water, nuts, and plants you know with certainty.
- Cook all wild meat, fish and shellfish thoroughly.

Best returns for little effort

1. Fishing with a line, hooks, or simple traps in streams
2. Insects and grubs
3. Familiar plants, nuts and berries you can identify with certainty
4. Trapping small animals (where it is legal and you know how)

See also: Foraging Plants Safely

Foraging Plants Safely

> Only eat a plant if you can identify it with **100% certainty**. Several deadly plants look like common edible ones.

Never eat

- **Wild mushrooms**, unless you are an expert. Some deadly species look harmless, and symptoms can take a day to appear, too late to treat.
- Plants that look like **wild carrot, parsnip or parsley** (white umbrella-shaped flower clusters). This family includes water hemlock and poison hemlock, among the deadliest plants anywhere.
- Plants with milky or discolored sap (some exceptions exist, but avoid unless you know them).
- Beans, bulbs or seeds inside pods, unless known.
- Plants with an almond smell (can mean cyanide).
- White and yellow berries (most are poisonous); about half of red berries are too.

Commonly known safe foods (where you know them)

- **Cattail:** young shoots, roots and pollen. Found in marshes.
- **Dandelion:** leaves, flowers and roots.
- **Clover:** leaves and flowers, best cooked.
- **Acorns:** must be shelled, crushed and soaked in several changes of water to leach out bitter tannins.
- **Pine:** inner bark and needles (tea). Avoid yew, which looks similar and is very toxic.
- **Blackberries, raspberries, blueberries:** aggregate and blue berries are among the safest.

The universal edibility test

Military manuals describe a slow test (touching, holding in the mouth, then eating small amounts over 24+ hours). It takes a long time, does not detect every toxin, and **does not work for mushrooms**. Use it only in true long-term emergencies with no other option.

Cooking helps

Boiling breaks down some toxins and kills parasites. Throw away the water if a plant is bitter.

Insects, Fishing and Small Game

Insects

Insects are high in protein and fat and easy to catch.

- **Good choices:** grasshoppers, crickets, ants, termites, beetle grubs (in rotten logs), earthworms.
- **Avoid:** brightly colored insects, hairy caterpillars, anything that stings or bites (except ants once cooked), flies, ticks and mosquitoes.
- Cook them: roast or boil. Remove legs and wings from grasshoppers. Cooking kills parasites.

Fishing

- **Improvised hooks:** bent safety pins, wire, thorns, carved bone or hardwood.
- **Line:** fishing line from your kit, inner strands of paracord, dental floss.
- **Bait:** insects, worms, small pieces of fish guts.
- Fish in the early morning and evening, in shade, near logs and in deeper pools below rapids.
- Set several lines at once and check them often, rather than holding one rod.
- A **fish trap**: a ring of stones in shallow water with a narrow funnel entrance. Fish swim in and struggle to find their way out.

Cook all freshwater fish thoroughly. Avoid fish that look sick, smell bad, or have slimy gills or sunken eyes.

Shellfish

- Freshwater mussels and crayfish are good, cooked well.
- Seashore shellfish can carry toxins from algae blooms (red tide), which cooking does not destroy. Avoid during warnings or in warm months if unsure.

Small game

Snares and traps work, but take skill and time, and are illegal in many places outside emergencies. Check many traps over a wide area. Always cook wild meat until it is no longer pink.

Keeping Food and Camp Safe

Keep animals away

- Cook and eat at least 60 m (200 ft) downwind from where you sleep.
- Hang food, trash and anything with a smell (toothpaste, soap) from a branch at least 4 m (12 ft) up and 1.5 m (5 ft) out from the trunk, or use a bear canister.
- Never keep food in your shelter in bear country.

Avoid food poisoning

- Wash hands with soap, or use sanitizer, before handling food.
- Boil water used for cooking and washing dishes.
- Eat cooked food right away. In warm weather, cooked food left out for more than 2 hours can make you sick.
- Throw away cans that bulge, leak, spurt when opened, or smell bad.

Preserving meat and fish

- **Drying:** cut into thin strips, no thicker than 6 mm (1/4 in), and dry in sun and wind or over a low, smoky fire until brittle.
- **Smoking:** keep strips in heavy smoke over a low fire for many hours. Smoke keeps insects off and helps preserve.
- **Cold:** in winter, freeze meat outside in a shaded, animal-proof place.

> Thin strips and complete drying are key. Thick or damp meat spoils quickly.

Emergency Food Storage

How much

At least **3 days** of food for everyone in the home. **Two weeks** is better. Include pets.

What to store

Choose food your family already eats, that needs little or no cooking and little water:

- Canned meat, fish, beans, soups, vegetables and fruit
- Peanut butter, nuts, trail mix
- Crackers, granola and energy bars
- Dried fruit
- Ready-to-eat cereal, instant oats
- Shelf-stable milk or powdered milk
- Rice, pasta and dried beans (only if you can cook with water to spare)
- Comfort food: chocolate, coffee, tea, sweets. Morale matters.
- Baby food and formula if you have an infant; special foods for anyone with allergies or medical needs

Do not forget

- A **manual can opener**
- Plates, cups and utensils you can use without washing
- A way to cook without power: camp stove and fuel, used **outdoors only**

Keeping it fresh

- Store in a cool, dry, dark place, off the floor.
- **First in, first out:** use the oldest food first and put new purchases at the back.
- Check dates twice a year (for example, when the clocks change).
- Throw out cans that are bulging, leaking, rusted deeply or dented along a seam.

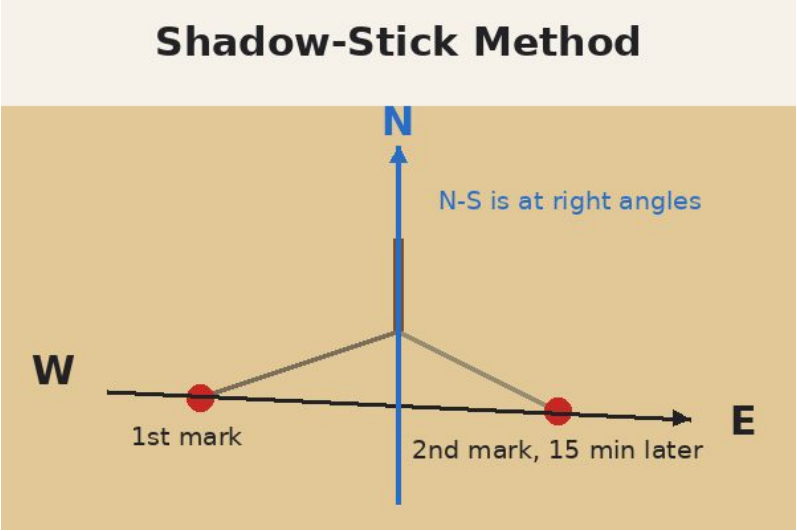
> Salty foods make you thirsty. If water is short, choose low-salt foods and canned foods packed in water or juice.

Navigation

Finding direction without a GPS

Shadow-Stick Method

This works anywhere in the world on a sunny day.



Shadow-stick method

Steps

1. Push a straight stick, about 1 m (3 ft) long, upright into flat ground.
2. Mark the tip of its shadow with a small stone. This is your **west** mark.
3. Wait **15 minutes** or more. The shadow tip moves.
4. Mark the new shadow tip. This is your **east** mark.
5. Draw a line from the first mark to the second: that line runs **west to east**.
6. Stand with the first mark on your left and the second on your right. You are now facing **north**.

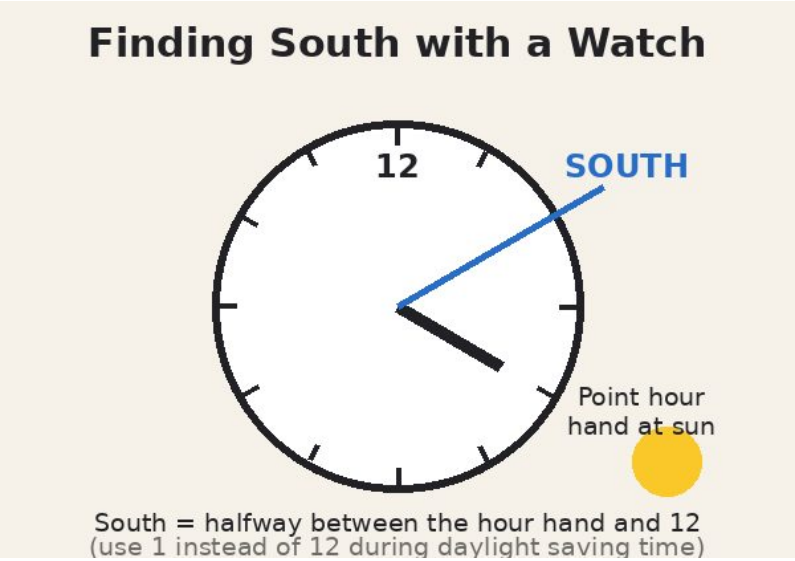
Tips

- The longer you wait between marks, the more accurate the result.
- It is most accurate around midday and near the equinoxes, and less accurate early and late in the day.

- At local noon, the shadow is at its shortest and points roughly north (in the Northern Hemisphere) or south (in the Southern Hemisphere), away from the sun.

See also: Finding South with a Watch

Finding Direction with a Watch



Watch method

Northern Hemisphere

1. Hold the watch flat.
2. Point the **hour hand** at the sun.
3. Halfway between the hour hand and the **12** is **south**.
4. During daylight saving time, use the **1** instead of the 12.

Southern Hemisphere

1. Point the **12** at the sun.
2. Halfway between the 12 and the hour hand is **north**.
3. During daylight saving time, use the 1 instead of the 12.

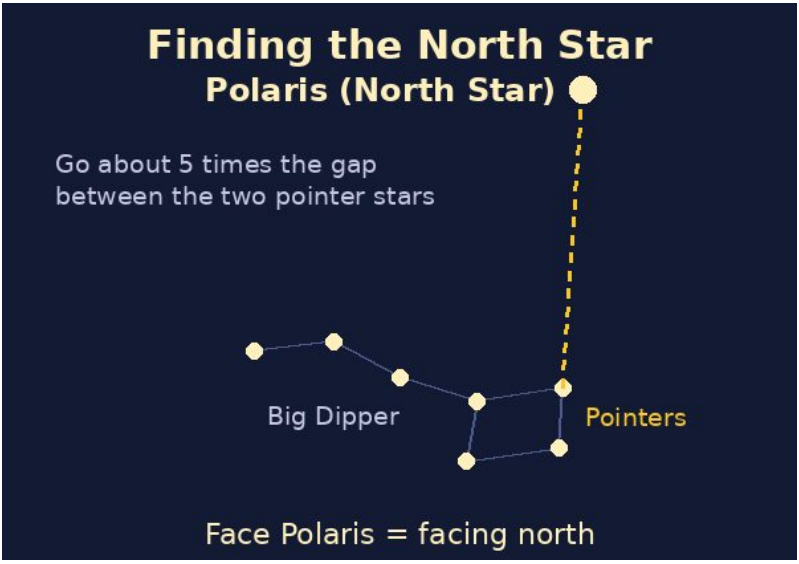
No analog watch?

Draw a clock face on paper or the ground with the hands showing the current time, and use it the same way.

> This method is a rough guide, off by 20 degrees or more in some places and times, and it does not work well near the equator. Check it against other methods.

Navigating by the Stars

Northern Hemisphere: the North Star

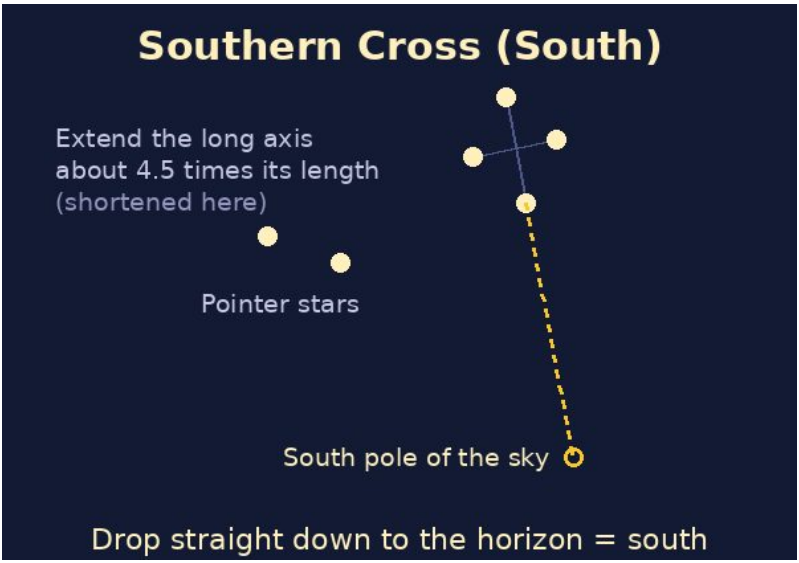


Finding the North Star

1. Find the **Big Dipper**, a ladle shape of seven bright stars.
2. The two stars on the outer edge of the bowl are the **pointers**.
3. Follow the line from the bottom pointer through the top pointer, about **5 times** the gap between them.
4. You reach **Polaris**, a moderately bright star on its own. It is the end of the handle of the Little Dipper.
5. Polaris is almost exactly over the North Pole. Facing it, you face **north**.

If the Big Dipper is low or hidden, look for **Cassiopeia**, a W shape on the opposite side of Polaris.

Southern Hemisphere: the Southern Cross



Southern Cross

1. Find the **Southern Cross**, four bright stars in a kite shape. Two bright **pointer stars** sit beside it.
2. Extend the long axis of the cross about **4.5 times** its length from the foot.
3. That point is roughly the **south pole of the sky**. Drop straight down to the horizon: that is **south**.

Orion (both hemispheres)

Orion's Belt, three bright stars in a row, rises almost exactly due **east** and sets almost due **west**.

Any star

Watch a star over two fixed sticks for 15 minutes. In the Northern Hemisphere: rising means you face east, sinking means west, moving right means south, moving left means north. Reverse north and south in the Southern Hemisphere.

Map and Compass Basics

Orient the map

1. Lay the compass on the map with its edge along a north-south grid line.
2. Turn the map and compass together until the red needle points to north on the map (allowing for declination, below).
3. The map now matches the land around you.

Take a bearing to a landmark

1. Point the travel arrow on the compass at the landmark.
2. Turn the dial until the orienting arrow lines up under the red needle ("red in the shed").
3. Read the bearing at the index line.

Follow a bearing

1. Hold the compass flat in front of you, travel arrow forward.
2. Turn your body until the needle sits in the orienting arrow.
3. Pick a tree or rock on that line, walk to it, then repeat.

Declination

Compass north (magnetic north) differs from map north. The difference, printed on most maps, can be 10-20 degrees or more in some regions. Adjust for it, or you will drift far off course over distance.

Other tips

- Keep metal, phones and magnets away from the compass.
- **Aiming off:** if heading to a point on a river or road, aim deliberately to one side, so you know which way to turn when you reach it.
- **Handrails:** follow long features like rivers, ridges or trails.
- **Count paces** to estimate distance: most people take about 60-70 double paces per 100 m on flat ground.

Travel and Staying Found

Before you go

Tell someone your route and when to expect you. Carry a map of the area.

Marking your trail

- Break branches at eye level, pointing your direction.
- Stack small rock piles (cairns) or make arrows with sticks.
- Leave notes at junctions: time, date, direction and condition.

If you are lost and decide to move

- Pick one direction and stick to it. People who wander in circles get more lost.
- Go downhill and follow water: streams lead to rivers, and rivers to roads and towns. But avoid steep canyons and waterfalls.
- Climb to a high point (safely) to look for roads, buildings, smoke or power lines.
- Follow power lines, fences, pipelines or railways. They lead somewhere.
- Travel in daylight. Stop and make camp well before dark.

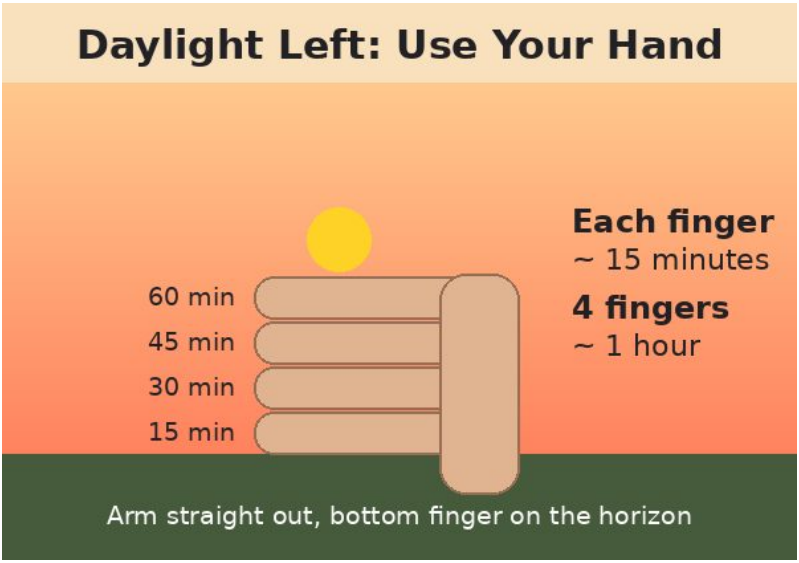
> In fog, darkness or a whiteout, stop and wait. Walking blind leads to falls and getting more lost.

Signs of people

Roads, trails, fences, cut stumps, tire tracks, livestock, and the sound of traffic, dogs or machines.

Natural Direction Clues and Daylight

How much daylight is left?



Estimating time until sunset with your hand

1. Stretch your arm out fully, palm toward you, fingers together and level.
2. Put your bottom finger on the horizon.
3. Count how many finger-widths fit between the horizon and the sun.
4. Each finger is roughly **15 minutes** of daylight; four fingers is about an hour.

This is a rough guide that changes with your location and the season. Leave plenty of extra time. After sunset, twilight gives about 20-30 more minutes of usable light in most places.

Sun basics

- The sun rises roughly in the **east** and sets roughly in the **west**.
- North of the tropics, the midday sun is in the **south**. South of the tropics, it is in the **north**.
- Sunrise and sunset are only exactly east and west near the equinoxes (March and September). In summer they shift toward the pole; in winter, toward the equator.

Clues that are often right (Northern Hemisphere)

Reverse north and south in the Southern Hemisphere.

- **Snow** melts first on south-facing slopes and lasts longer on north-facing ones.
- South-facing slopes are often drier, with more open, sun-loving plants. North-facing slopes are often damper with denser growth.
- **Prevailing winds** shape trees and snowdrifts. If you know the usual wind direction in your area, bent trees and drift shapes point the way.
- **Churches:** older churches in Europe often have the altar at the east end.
- **Satellite dishes** in North America and Europe tend to face southward.

Myths - do not rely on them

- **"Moss grows on the north side of trees."** Moss grows wherever it is damp and shady, which can be any side.
- **"Ant hills are on the south side."** Not reliable.
- **Tree rings wider on one side** depend on slope, light and wind, not direction.

> Combine several clues and confirm with the Shadow-Stick Method or the stars before committing to a direction.

Signals & Rescue

Getting found and calling for help

Ground-to-Air Signals

Large symbols on the ground can be seen from aircraft kilometers away.



Ground-to-air signals

The key symbols

- **V** - Need assistance
- **X** - Need medical assistance
- **N** - No
- **Y** - Yes
- **Arrow** - Going in this direction
- **SOS** - Distress (understood everywhere)

Making them visible

- Make each symbol at least **3 m (10 ft)** long, with lines about 1 m (3 ft) wide. Bigger is better.

- Use **contrast**: dark logs or rocks on snow, light stones or clothing on dark ground, trench lines in snow or sand so shadows show.
- Make straight lines and sharp angles. Nature has few of them, so they stand out.
- Pick an open area: a clearing, beach, ridge or frozen lake.

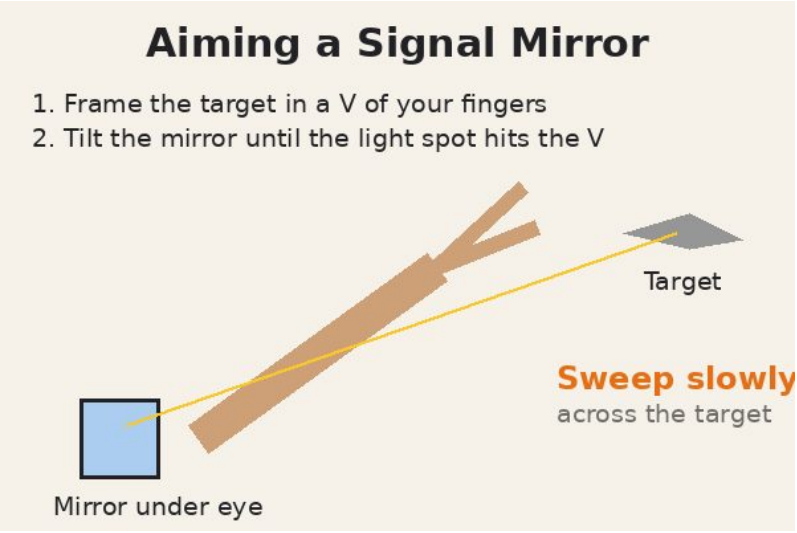
Answering aircraft with your body

- **Both arms raised in a Y:** Yes, we need help.
- **One arm up, one arm down (diagonal N):** No, we do not need help.

> If an aircraft rocks its wings or flashes lights, it has seen you.

Signal Mirror

A mirror flash can be seen up to 15 km (10 miles) or more on a sunny day. It is one of the best rescue signals.



Aiming a signal mirror

Aiming without a sighting hole

1 Hold the mirror just under your eye, shiny side out.

2 Stretch out your other arm and make a **V** with two fingers. Frame the target (aircraft, boat, hilltop) inside the V.

3 Tilt the mirror until the spot of sunlight lands on your fingers.

4 Keep the light in the V and rock the mirror slightly so it sweeps across the target.

Mirrors with a sighting hole

Look through the hole at the target. You will see a bright aiming spot; tilt the mirror until the spot sits on the target.

Tips

- Anything shiny works: a phone screen, CD, polished can lid, foil.
- Sweep the horizon often, even when you see nothing. Pilots may see you before you see them.
- Do not keep flashing a pilot who has clearly seen you; it can blind them. Switch to ground signals.

SOS, Whistle, Fire and Smoke

The rule of three

Three of anything means distress: three fires in a triangle, three whistle blasts, three gunshots, three flashes.

SOS in Morse code



SOS in Morse code

Three short, three long, three short: `...` `---` `...`. Pause, then repeat. Use a flashlight, mirror, whistle or car horn.

Whistle

A whistle carries much farther than shouting and uses less energy. Blow **three long blasts**, then pause and listen. Repeat regularly. The answer is often two blasts.

Signal fires

- Build three fires in a triangle, about 30 m (100 ft) apart, in an open area.
- Keep them ready to light: tinder and kindling under cover, with plenty of fuel nearby.
- **Day:** add green leaves, moss or wet wood to make thick **white smoke**, which shows against dark forest. Rubber or oil makes **black smoke**, which shows against snow or sand.
- **Night:** bright flames show best.

> Only light a signal fire when you hear or see a search aircraft or party, unless you have endless fuel. Never risk starting a wildfire.

Other signals

- Bright clothing or a space blanket spread out in the open.
- Flashlight or headlamp at night, in SOS pattern.

- Move: motion is easier to spot than a still person.

Radio and Phone Tips

Mobile phones

- Try calling your emergency number (911, 112) even with no bars. Emergency calls can use any carrier's tower.
- **Text messages** need less signal than calls. Many regions accept texts to 911 or 112.
- Climb high or move to open ground for a better signal.
- Save battery: airplane mode, low brightness, and check for signal at set times.
- Some newer phones can send emergency messages by satellite. Check your phone's emergency settings before a trip.

Common emergency radio channels (North America)

- **CB:** channel 9 (emergency), channel 19 (truckers)
- **FRS/GMRS:** channel 1 or 3 are commonly monitored in some areas. No license needed for FRS.
- **Amateur (ham) 2 m:** 146.520 MHz (national calling frequency). A license is normally required to transmit, but anyone may transmit in a genuine emergency.
- **Marine VHF:** channel 16 (156.8 MHz), distress and calling
- **NOAA Weather Radio:** 162.400-162.550 MHz (receive only)

> Channel use differs by country. In Europe, PMR446 radios are common, and 112 works on most phones.

Making a distress call

Say **"Mayday, Mayday, Mayday"** (life-threatening) or **"Pan-pan"** (urgent), then: who you are, where you are, what happened, how many people, and what help you need. Repeat, then listen.

Personal locator beacons (PLB)

A PLB or satellite messenger sends your position to rescuers by satellite, anywhere in the world with a view of the sky. It is the most reliable way to call for help in remote areas.

Signaling at Night

At night, even a small light can be seen from far away, especially by aircraft. Night-vision equipment used by many rescue crews makes lights even easier to spot.

Flashlight or headlamp

- Flash **SOS**: three short, three long, three short. See SOS, Whistle, Fire and Smoke.
- Point it at the aircraft or toward searchers' lights. Sweep slowly across them.
- Save batteries: do not leave it on constantly. Signal when you hear or see searchers, and on a schedule (for example, for a minute at the top of every hour).

Glow stick on a cord

Tie a glow stick to 1 m (3 ft) of cord and swing it in a big circle. It makes a large, moving ring of light that stands out from the ground.

Phone

- Use the flashlight function to signal.
- A bright white screen is visible surprisingly far in the dark.
- Turn the phone off between signals to save battery.

Fire

Bright, tall flames show best at night. Keep a signal fire built and ready to light, with fast-burning fuel on top. See SOS, Whistle, Fire and Smoke.

Reflectors

Hang anything reflective (space blankets, reflective strips on packs and jackets) around camp. A searcher's flashlight or spotlight will make it shine.

Sound travels further at night

Air is often still and quiet at night. Blow your whistle in sets of three and then listen carefully for a reply.

Helping Searchers Find You

Search teams work in a systematic way. Help them by staying put, being visible, and making noise.

Stay in one place

Searchers look hardest around your last known location and route. A moving person is much harder to find, and may walk out of an area that was already searched.

Be easy to see

- Go to the most open spot nearby: a clearing, ridge, meadow, beach or riverbank.
- Spread bright clothing, tarps or space blankets in the open.
- Build Ground-to-Air Signals and keep a signal fire ready.
- Use a Signal Mirror on any aircraft you see.

Be easy to hear

- Blow your whistle in sets of three, then pause and listen.
- Bang metal or rocks if you have no whistle.
- **Answer** when you hear your name called. Some lost people hide from searchers out of fear or embarrassment. Do not.

Leave clues if you move

If you must move, leave a note at your last spot with the time, your direction and your condition. Mark your route with arrows and broken branches.

When a helicopter comes

- Stay in the open and wave both arms in a **Y** (yes, we need help).
- Do not approach a helicopter unless the crew signals you to. Come from the **front**, where the pilot can see you, crouching low.

- Never go near the tail rotor. Never approach from uphill, where blades are closer to the ground.
- Tie down or hold onto loose items like tarps, hats and sleeping bags; the downdraft is very strong.
- If you can, clear a landing area about 30 x 30 m (100 x 100 ft), flat, with no wires, tall trees or loose debris.

When searchers reach you

Tell them: how many people, injuries, medical conditions, and anyone still missing.